

# STEMS: Safe Multi-Agent Reinforcement Learning for Buildings, Heat Pumps and Electric Vehicles

Replication, Mathematics, and Experimental Record on Real CityLearn Data

September 2, 2026

## Abstract

This report is a complete record of a from-scratch reimplementation of STEMS (Spatial-Temporal Enhanced Multi-Agent Safe Building Energy Management System) [1] and of the extensions built on top of it. It documents, in order: the real CityLearn environments used and every device equation they expose; the full mathematical formulation of the architecture, from the building-similarity graph through the spatial-temporal encoder to the constrained actor-critic update; a unified control-barrier framework covering the battery, the hot-water tank and electric-vehicle charging as instances of one object; the evaluation metrics; and each experiment that was run – including a sweep of a shared grid constraint across a fleet of deadline-constrained vehicles – with its measurements, its limitations, and – where a result was negative or a mechanism inactive – the specific work planned to address it. A dedicated section describes standalone operation of the heat-pump and hot-water controller and sets out what would be required to run it on a commercial heat-pump platform. All measurements reported here were taken on the real simulator; anything synthesised or derived is marked as such where it appears.

## Contents

|          |                                                                 |          |
|----------|-----------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction and Scope</b>                                   | <b>3</b> |
| 1.1      | What this report covers . . . . .                               | 3        |
| 1.2      | Reporting conventions . . . . .                                 | 3        |
| <b>2</b> | <b>The CityLearn Environments</b>                               | <b>3</b> |
| 2.1      | The building schema . . . . .                                   | 4        |
| 2.2      | Device equations . . . . .                                      | 4        |
| 2.3      | An environment containing both buildings and vehicles . . . . . | 5        |
| 2.4      | Heterogeneous buildings . . . . .                               | 5        |
| <b>3</b> | <b>Architecture and Mathematical Formulation</b>                | <b>6</b> |
| 3.1      | Building similarity graph . . . . .                             | 6        |
| 3.2      | Spatial encoder . . . . .                                       | 6        |
| 3.3      | Temporal encoder . . . . .                                      | 6        |
| 3.4      | Fusion and per-building actor-critic . . . . .                  | 6        |
| 3.5      | Constrained update . . . . .                                    | 7        |

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| <b>4</b>  | <b>Safety: One Barrier Framework, Four Devices</b>                 | <b>7</b>  |
| 4.1       | Classical barriers . . . . .                                       | 7         |
| 4.2       | Lagrangian safety pressure . . . . .                               | 7         |
| 4.3       | The unification: deadline-constrained storage . . . . .            | 7         |
| 4.4       | Hot water: estimating the requirement . . . . .                    | 8         |
| 4.5       | Weather-dependent efficiency . . . . .                             | 9         |
| 4.6       | Coupled feasibility . . . . .                                      | 9         |
| <b>5</b>  | <b>Reward and Metrics</b>                                          | <b>9</b>  |
| <b>6</b>  | <b>Experiments</b>                                                 | <b>10</b> |
| 6.1       | E1 — Battery calibration and the safety ablation . . . . .         | 10        |
| 6.2       | E2 — Converged full-year training run . . . . .                    | 10        |
| 6.3       | E3 — Hot-water lead time . . . . .                                 | 11        |
| 6.4       | E4 — Anticipatory pre-heating . . . . .                            | 11        |
| 6.5       | E5 — Cold-snap stress . . . . .                                    | 12        |
| 6.6       | E6 — Vehicle barrier on real charger data . . . . .                | 14        |
| 6.7       | E7 — The merged building-and-vehicle environment . . . . .         | 14        |
| 6.8       | E8 — Regression verification . . . . .                             | 14        |
| 6.9       | E9 — Coupling sweep with a deadline-locked vehicle fleet . . . . . | 14        |
| <b>7</b>  | <b>Standalone Heat-Pump Operation and Platform Integration</b>     | <b>17</b> |
| 7.1       | Running standalone . . . . .                                       | 17        |
| 7.2       | What the controller needs from a heat-pump installation . . . . .  | 17        |
| 7.3       | Integration path for a commercial platform . . . . .               | 17        |
| 7.4       | Recalibration required for a different market . . . . .            | 18        |
| <b>8</b>  | <b>Defects Found and Corrected</b>                                 | <b>19</b> |
| <b>9</b>  | <b>Limitations and Planned Work</b>                                | <b>19</b> |
| <b>10</b> | <b>Summary</b>                                                     | <b>21</b> |

# 1 Introduction and Scope

Buildings account for a large and increasingly electrified share of grid demand. Batteries, heat pumps, hot-water stores and electric vehicles are all controllable, all constrained, and all coupled through the same electrical connection. Coordinating them is a constrained sequential decision problem: an agent must reduce cost, peak demand and emissions while keeping every device inside its physical limits and keeping occupants comfortable. Safe reinforcement learning – RL augmented with an explicit safety layer – is the approach taken by the paper this work reimplements [1].

## 1.1 What this report covers

The work proceeded in four stages, and this report follows that order:

1. **Replication on real data.** A faithful reimplementation of the STEMS architecture, run on a real eight-building CityLearn simulation, with every device parameter read from the simulator rather than assumed (Sections 2–5).
2. **Calibration and safety experiments** on the battery-side control barrier (Section 6.1).
3. **Thermal extension** – hot water and the heat pump, including anticipatory pre-heating and weather-dependent efficiency (Sections 6.4–6.5), with standalone operation and platform integration treated separately in Section 7.
4. **Electric vehicles** – unifying the hot-water and vehicle problems under one barrier, and constructing an environment in which buildings and vehicles coexist (Sections 6.6–6.7).

## 1.2 Reporting conventions

Three conventions are used throughout, and are worth stating because they shape how the numbers should be read.

- **Measured, not assumed.** Every device rate, capacity and efficiency used by the controller is read from the live simulator object. Where a quantity is estimated rather than measured, the estimator is described and its inputs are restricted to information available at that timestep.
- **Fail loud.** A missing observation, an absent device or an unsupported configuration raises an error rather than resolving silently to a default. Simulator corrections are announced and stamped into the run metadata.
- **Negative results are reported in place.** Where a mechanism had no effect, or improved one quantity at the expense of another, the measurement is given as observed and followed by the specific planned work (Section 9).

# 2 The CityLearn Environments

CityLearn (2.6.0b1) is an open-source Gymnasium building-energy simulator. It exposes per-building observations and actions and advances the simulation using physically motivated device models driven by historical weather and load data.

## 2.1 The building schema

| Parameter            | Value                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------|
| Source               | NREL ResStock AMY2018–2021, Travis County, Texas                                                        |
| Buildings simulated  | 8 (of 100 candidates), decentralised                                                                    |
| Horizon              | steps 0–8759 at 3600 s $\Rightarrow$ one full year, hourly                                              |
| Devices per building | bidirectional heat pump (cooling + heating),<br>electric DHW heater, battery, DHW storage tank, PV      |
| Sizing               | <code>autosize: true</code> ; devices sized at <code>reset()</code>                                     |
| Observations         | 45 active; 28 selected by name, +2 in heat-pump mode                                                    |
| Actions              | [ <code>dhw_storage</code> , <code>electrical_storage</code> , <code>cooling_or_heating_device</code> ] |

Observations are mapped from CityLearn’s names to the STEMS subset *by name*, so the selection is explicit and auditable rather than positional.

## 2.2 Device equations

**Storage (battery and DHW tank).** Both inherit the same base dynamics. For requested energy  $e$  (positive = charge):

$$\text{SOC}^{\text{init}} = \max(0, \text{SOC}_{t-1} C (1 - \lambda)), \quad (1)$$

$$\text{SOC}_t = \frac{1}{C} \begin{cases} \min(\text{SOC}^{\text{init}} + e\sqrt{\eta}, C) & e \geq 0, \\ \max(0, \text{SOC}^{\text{init}} + e/\sqrt{\eta}) & e < 0, \end{cases} \quad (2)$$

with capacity  $C$ , round-trip efficiency  $\eta$  and standby loss  $\lambda$ .

**Heat pump.** A temperature-dependent Carnot coefficient of performance is computed for each mode and clipped to  $[0, 20]$ :

$$\text{CoP}_{\text{heat}}(T_{\text{out}}) = \eta \frac{T_{\text{tgt}}^{\text{heat}} + 273.15}{T_{\text{tgt}}^{\text{heat}} - T_{\text{out}}}, \quad \text{CoP}_{\text{cool}}(T_{\text{out}}) = \eta \frac{T_{\text{tgt}}^{\text{cool}} + 273.15}{T_{\text{out}} - T_{\text{tgt}}^{\text{cool}}}, \quad (3)$$

$$P_{\text{out}} = \min(|a| P_{\text{nom}}, P_{\text{nom}}) \cdot \text{CoP}, \quad P_{\text{in}} = P_{\text{out}}/\text{CoP}. \quad (4)$$

Which mode is active is *not* selected by the sign of the action: an exogenous, data-derived `hvac_mode` schedule gates it, and the scalar action scales whichever device that schedule has enabled. This is why the comfort term must be season-aware (Section 5).

**DHW heater.** A resistive model,  $P_{\text{out}} = \min(|a| P_{\text{nom}}, P_{\text{nom}}) \eta$  with  $\eta \approx 0.9\text{--}0.99$ , charging the tank through Eqs. 1–2.

**PV.** Generation is read from a PySAM PVWatts simulation executed once at `reset()`, constrained by each building’s roof area.

CityLearn ships a default reward; STEMS does not use it and computes its own (Section 5).

## 2.3 An environment containing both buildings and vehicles

Neither shipped dataset expresses the full problem. The Travis schema has no chargers. The EV challenge dataset has chargers but *no thermal observations at all* – no indoor temperature, setpoints, hot water, occupancy or cooling demand – so it cannot represent the building side.

`setup_citylearn_ev.py` constructs a merged schema, `tx.travis_8b_ev`: the Travis buildings unchanged, plus charger hardware, vehicle battery definitions and per-charger schedules.

|                       |                                                  |
|-----------------------|--------------------------------------------------|
| Buildings             | 8, of which <b>6 have a charger</b>              |
| Observation dimension | 35 = 28 base + 2 heat-pump + 5 EV                |
| Action dimension      | 4: DHW, battery, heat pump, EV charger           |
| Charger hardware      | 11 kW / 7.4 kW, $\eta = 0.95$ , 7.2 kW discharge |
| Vehicle batteries     | 40–75 kWh                                        |
| Charging lead time    | <b>5.7–7.4 h</b>                                 |

No building data is copied. CityLearn resolves a charger file as `os.path.join(root_directory, charger_simulation)`, and `os.path.join` returns an absolute right-hand side unchanged, so the charger schedules live in the project directory while every building file continues to resolve from the untouched simulator cache. The thermal side of the merged schema is therefore identical to the original.

Partial penetration (6 of 8) is deliberate: it leaves two buildings that can only *offer* flexibility and never demand it.

**Provenance.** The building thermal models, loads, weather, pricing and carbon intensity are real and unmodified. The **vehicle schedules are synthetic** – ResStock carries no vehicle data – generated from a seeded, documented commuter model: weekday departure  $\sim 07:30$  and return  $\sim 17:30$ , weekend 09:30/15:30, stay-at-home probability 12%/45%, required state of charge at departure  $\mathcal{U}(0.70, 0.90)$ , arrival state of charge  $\mathcal{U}(0.25, 0.55)$  decaying with time away. Every result measured on this schema carries that caveat.

## 2.4 Heterogeneous buildings

CityLearn buildings need not agree with one another: in the EV challenge data the native action dimension is 1, 2 or 3 and the observation dimension 28, 35, 37 or 42, because only some buildings own a charger and charger identifiers are embedded in observation names.

The architecture requires one shape for all buildings, since the graph network stacks them into a single matrix. The wrapper therefore builds a *canonical padded layout* together with a per-building map into native indices:

- **Observations** = base features +  $5 \times$  (EV slots). A building with fewer bays is zero-padded, and a padded slot reports `connected_state` = 0, which the vehicle barrier already interprets as "no vehicle". An absent charger and an empty bay therefore follow the same code path.
- **Actions** follow a canonical order restricted to devices some building owns. Slots a building lacks are *removed* before the vector reaches the simulator, because CityLearn sizes its action array to the devices that exist.

On the homogeneous building schema every map is the identity, so this reduces exactly to the original behaviour – verified by test and by re-running a full study (Section 6.8).

### 3 Architecture and Mathematical Formulation

Per timestep, for each building  $i$ :

$$o_i \xrightarrow{\text{normalise}} \underbrace{\text{GCN} \oplus \text{Transformer}}_{\text{shared encoder}} \rightarrow r_i \in \mathbb{R}^{64} \xrightarrow{\pi_\theta} a_i^{\text{raw}} \xrightarrow{\text{barrier}} a_i^{\text{safe}} \rightarrow \text{CityLearn}.$$

#### 3.1 Building similarity graph

Buildings are connected by a graph combining geographic proximity and functional similarity:

$$w_{ij} = \alpha \exp\left(-\frac{d_{ij}^2}{2\sigma_d^2}\right) + \beta \exp\left(-\frac{\|f_i - f_j\|^2}{2\sigma_f^2}\right), \quad w_{ii} = 0, \quad (5)$$

with  $d_{ij}$  the geographic distance and  $f_i$  a normalised functional feature vector (battery capacity, floor area).

#### 3.2 Spatial encoder

A three-layer graph convolutional network propagates information across the  $B$  buildings using the symmetrically normalised adjacency  $\hat{A} = A + I$ :

$$H^{(l+1)} = \sigma\left(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)}\right), \quad \hat{D}_{ii} = \sum_j \hat{A}_{ij}, \quad (6)$$

with  $\sigma = \text{ReLU}$  and hidden width 64.

#### 3.3 Temporal encoder

A multi-head self-attention block attends over a rolling  $T = 24$  h window of each building's own history:

$$e_{\tau, \tau'} = \frac{Q_\tau K_{\tau'}^\top}{\sqrt{d_k}}, \quad z_{i,t} = \text{LayerNorm}\left(x_{i,t} + \sum_{\tau'} \text{softmax}_{\tau'}(e_{t, \tau'}) V_{\tau'}\right), \quad (7)$$

with 4 heads, a 32-dimensional embedding and fixed sinusoidal positional encodings.

#### 3.4 Fusion and per-building actor-critic

$$r_i = W_s h_i + W_t z_i + b. \quad (8)$$

Each building has its own squashed-Gaussian actor, value critic and multi-head cost critic conditioned on  $r_i$ :

$$z \sim \mathcal{N}(\mu(r_i), \sigma(r_i)^2), \quad a = \tanh(z), \quad (9)$$

$$\log \pi_\theta(a \mid r_i) = \log \mathcal{N}(z; \mu, \sigma^2) - \sum_d \log(1 - \tanh^2(z_d) + \epsilon), \quad (10)$$

$$V_\phi(r_i) \in \mathbb{R}, \quad V_\psi^c(r_i) \in \mathbb{R}^K \quad (K = 3). \quad (11)$$

### 3.5 Constrained update

Training is on-policy, one update per full-year episode. Critics are trained by TD regression against target networks; the actor by a generalised advantage estimate ( $\gamma = 0.99$ ,  $\lambda = 0.95$ ) penalised by the Lagrangian cost advantage, with an auto-tuned entropy temperature (target entropy  $-|\mathcal{A}|$ ):

$$\mathcal{L}_{\text{critic}} = \mathbb{E}[(V_\phi(r_i) - (r + \gamma V_\phi^{\text{tgt}}(r'_i)))^2], \quad (12)$$

$$\mathcal{L}_{\text{cost}} = \mathbb{E}[(V_\psi^c(r_i) - (c + \gamma V_\psi^{c,\text{tgt}}(r'_i)))^2], \quad (13)$$

$$\hat{A}_i^{\text{eff}} = \hat{A}_i^{\text{GAE}} - \sum_k \max(\lambda_k, 0) \hat{A}_{i,k}^c, \quad (14)$$

$$\mathcal{L}_{\text{actor}} = -\mathbb{E}[\hat{A}_i^{\text{eff}} \log \pi_\theta(a_i | r_i)] + \alpha \mathbb{E}[\log \pi_\theta(a_i | r_i)]. \quad (15)$$

## 4 Safety: One Barrier Framework, Four Devices

### 4.1 Classical barriers

Three barriers are evaluated on the *next predicted* state:

$$h_1 : \text{SOC} \in [\text{SOC}_{\min}, \text{SOC}_{\max}] = [0.1, 0.9], \quad (16)$$

$$h_2 : |e_i| \leq P_{\text{building,max}}, \quad (17)$$

$$h_3 : \sum_i \max(e_i, 0) \leq P_{\text{grid,max}}. \quad (18)$$

Because  $h_1$  is decoupled across buildings it is solved *analytically* per building rather than through a general quadratic program – exact, and fast enough for the  $8760 \times 8$  step budget. With the per-building state-of-charge change per unit action  $\rho_i = P_{\text{nom},i}/C_i$  read from the real battery, a robust margin  $m$  and an anticipatory horizon  $H$ :

$$[\ell_i, u_i] = \left[ \text{SOC}_{\min} + m + \frac{1}{2}\rho_i H, \text{SOC}_{\max} - m - \frac{1}{2}\rho_i H \right], \quad (19)$$

$$a_i^{\text{batt,safe}} = \begin{cases} \text{clip}\left(a_i^{\text{nom}}, \frac{\ell_i - \text{SOC}_i}{\rho_i}, \frac{u_i - \text{SOC}_i}{\rho_i}\right) & \text{if feasible,} \\ +1 & \text{if } \text{SOC}_i < \ell_i \text{ (recover: charge),} \\ -1 & \text{if } \text{SOC}_i > u_i \text{ (recover: discharge).} \end{cases} \quad (20)$$

The projection is feasible by construction: a recovery action exists even when the state is already outside the band, so the shield never falls back to an all-zero emergency action.

### 4.2 Lagrangian safety pressure

Alongside the hard shield, a soft penalty trains the *policy* to be safe, with one multiplier per constraint. With cost-rate error  $\epsilon_k = J_{c_k} - d$  and target rate  $d$ :

$$I_k \leftarrow \text{clip}(I_k + k_i \epsilon_k, 0, \lambda_{\max}), \quad \lambda_k \leftarrow \text{clip}(k_p \epsilon_k + I_k + k_d \text{relu}(J_{c_k}^{(t)} - J_{c_k}^{(t-1)}), 0, \lambda_{\max}). \quad (21)$$

### 4.3 The unification: deadline-constrained storage

Three of the four controllable devices are the same mathematical object:

*a store that must reach a required state of charge by a deadline, while charging at a finite rate.*

Table 1: The same barrier, three devices.

|                    | DHW tank              | Electric vehicle                    | Battery             |
|--------------------|-----------------------|-------------------------------------|---------------------|
| Deadline           | implicit: horizon $L$ | <b>declared</b>                     | none (a band)       |
| Requirement        | forecast demand / $C$ | <b>declared</b>                     | $\text{SOC}_{\min}$ |
| Capacity           | fixed                 | <b>varies per arrival</b>           | fixed               |
| Rate $\rho$        | $\min(c, P\eta/C)$    | $P_{\text{ch}}\eta/C_{\text{batt}}$ | $P_{\text{nom}}/C$  |
| Measured lead time | 1.2–1.9 h             | 5.7–7.4 h                           | —                   |
| Slack              | always 0              | <b>positive</b>                     | —                   |

**DeadlineStorageBarrier** implements the shared mechanism once; the hot-water and vehicle barriers are subclasses supplying only a requirement function. This is a discrete-time, rate-limited instance of input-constrained control-barrier methods, in which safety under actuator limits is certified by predicting forward under a backup control (here, "charge at maximum") over a finite horizon.

**Horizon rule.** With gap  $g = s^{\text{req}} - s$  and rate  $\rho$ ,

$$\text{steps\_needed} = \lceil g/\rho \rceil, \quad \text{slack} = \text{steps\_to\_deadline} - \text{steps\_needed}, \quad (22)$$

and  $\text{slack} \leq 0$  is a *latest-start* trigger: charge now or miss. The hot-water tank passes  $\text{steps\_to\_deadline} = 0$ , so every unmet requirement is immediately urgent; a vehicle passes its real time to departure, which is what creates deferral room.

**Projection.** Monotone – it raises the action and never lowers one, so it can add readiness but can never veto a policy that already charges harder:

$$a_i^{\text{safe}} = \max \left( a_i^{\text{nom}}, \min \left( c_i, \frac{\max(s_i^{\text{req}} - s_i, 0)}{\rho_i} c_i \right) \right). \quad (23)$$

#### 4.4 Hot water: estimating the requirement

CityLearn exposes hot-water demand for the current step only; there is no demand forecast observation. The requirement is therefore estimated online as an hour-of-day climatology over already-observed steps, per building  $i$ , hour  $h$  and weekday/weekend flag  $w$ :

$$m_{i,h,w} \leftarrow m_{i,h,w} + \alpha (d_{i,t} - m_{i,h,w}), \quad (24)$$

$$\hat{D}_i(L) = \sum_{k=1}^L m_{i,(h+k) \bmod 24, w} (1 + g \text{relu}(\bar{T} - \hat{T}_{t+k})), \quad (25)$$

with a cold-weather uplift. No future information enters:  $\hat{T}_{t+k}$  comes from the 1/2/3-step outdoor temperature predictions already present in the observation vector, and  $m$  is built only from past steps. During warm-up the estimator falls back to  $L \cdot d_{i,t}$ , which never reports less demand than is occurring. The barrier is

$$h_4 : \quad \text{SOC}_i^{\text{dhw}} \geq \text{clip} \left( \frac{\hat{D}_i(L)}{C_i} + m_{\text{dhw}} + \kappa \delta_i^{\text{cop}}, 0, s_{\max} \right). \quad (26)$$

## 4.5 Weather-dependent efficiency

Weather enters twice.

**Efficiency-aware power barriers.** Space conditioning draws  $|a|P_{\text{nom}}$  electrical, and the thermal service that buys scales with the coefficient of performance, so the same comfort costs more power on a cold hour. The guard counts that draw and derates the enforced cap in proportion to the efficiency shortfall against a rated reference:

$$P_{\text{max},i}^{\text{eff}} = P_{\text{max}} \left( 1 - \text{clip} \left( 1 - \frac{\text{CoP}_i(T_{\text{out}})}{\text{CoP}_i(T_{\text{ref}})}, 0, 0.5 \right) \right). \quad (27)$$

**Cold-front anticipation.** A falling temperature forecast means both higher upcoming demand and lower efficiency, so it is worth storing heat before the front arrives:

$$\delta_i^{\text{cop}} = \text{clip} \left( \frac{\text{CoP}_i(T_t) - \min_{k \leq H} \text{CoP}_i(\hat{T}_{t+k})}{\text{CoP}_i(T_t)}, 0, 1 \right), \quad (28)$$

which is identically zero under steady or improving weather.

## 4.6 Coupled feasibility

Each barrier is individually feasibility-guaranteed. That property does not survive coupling. Under a shared power cap, meeting every deadline requires

$$\sum_i \frac{(s_i^{\text{req}} - s_i) C_i}{\eta_i} \leq P_{\text{cap}} \Delta t T, \quad (29)$$

with  $T$  the time to the earliest binding deadline. When this fails the safe set is *empty*: no action sequence satisfies every deadline and the cap simultaneously, and no per-device projection can repair it.

The implementation detects the condition rather than absorbing it. `coupled_feasibility()` evaluates Eq. 29; `prioritise()` allocates the available power earliest-deadline-first, so that when something must slip the choice is an explicit policy rather than an artifact of array ordering; and `CBFShield.feasibility_report()` surfaces both. Reporting an unreachable requirement is the alternative to claiming a guarantee that does not hold.

## 5 Reward and Metrics

The reward decomposes additively into four terms per building per timestep:

$$r_i^{\text{econ}} = -\mu v_t e_i, \quad (30)$$

$$r_i^{\text{stab}} = \alpha_g \left( 1 - \frac{E_{\text{grid}}}{P_{\text{grid,max}}} \right)^2 + \alpha_b \left( 1 - \frac{|e_i|}{P_{\text{building,max}}} \right) - \beta_r \frac{|e_i - e_i^{\text{prev}}|}{P_{\text{building,max}}}, \quad (31)$$

$$r_i^{\text{comfort}} = -\lambda_c \cdot d(T_i^{\text{in}})^2, \quad d(T^{\text{in}}) = \begin{cases} T^{\text{in}} - T^{\text{cool}} & T^{\text{in}} > T^{\text{cool}} \\ T^{\text{heat}} - T^{\text{in}} & T^{\text{in}} < T^{\text{heat}} \\ 0 & \text{otherwise} \end{cases}, \quad (32)$$

$$r_i^{\text{renew}} = \xi \cdot \min \left( \frac{\text{solar}_i}{\max(\text{load}_i, \epsilon)}, 1 \right). \quad (33)$$

The dual-setpoint deadband in Eq. 32 is a correctness requirement rather than a refinement: with a single cooling setpoint the agent is penalised for being comfortably warm in winter. Any control mode that drives the heat pump enables it automatically.

**Metrics.** Seven quantities are computed from *observed* post-step state, never from the barrier’s internal model, so the safety figure cannot be influenced by the shield’s own assumptions: cost, emissions, average daily peak, total consumption, ramping rate, discomfort rate, and safety violation rate. Metrics 1–5 are reported normalised against a rule-based reference controller, following the convention of the metric definition; 6–7 are absolute. Violations are additionally decomposed into *avoidable* and *unavoidable* from the pre-action state, which distinguishes a control failure from a physically forced one. Hot-water experiments add readiness rate, coverage rate, energy deficit and mean tank state of charge.

## 6 Experiments

### 6.1 E1 — Battery calibration and the safety ablation

The dominant factor in the measured safety-violation rate is how precisely the barrier’s state-of-charge model matches each building’s actual battery. Reading the real per-building dynamics gives  $\rho_i = P_{\text{nom},i}/C_i \in [0.178, 0.533]$  – a threefold variation across eight buildings. An earlier implementation used a single constant 0.1 for every building.

The same seeded nominal policy was replayed through the real environment under different safety configurations, which isolates the shield’s marginal effect from training noise.

Table 2: E1: safety configuration ablation (800 steps  $\times$  2 seeds).

| Safety configuration                    | Violation rate | Cost        |
|-----------------------------------------|----------------|-------------|
| No barrier (raw policy)                 | 0.7195         | 7676        |
| Barrier, single uniform rate estimate   | 0.0945         | 7777        |
| Barrier + real per-building calibration | 0.0110         | 7703        |
| + robust margins                        | 0.0000         | 7705        |
| + anticipatory buffer                   | 0.0000         | 7662        |
| <b>Full configuration</b>               | <b>0.0000</b>  | <b>7649</b> |

Three observations. Calibration precision accounts for most of the improvement: a uniform rate estimate leaves a 9.45% violation rate even with the shield active, because it under-projects the safe action relative to what each battery actually does. Per-building calibration alone reaches 0.0110. The full configuration reaches zero measured violations and simultaneously the lowest cost, because on this dataset the power and grid limits never bind – loads are 10–40 kW against far larger caps – so the state-of-charge barrier is the only active constraint and it is now solved exactly.

### 6.2 E2 — Converged full-year training run

A 15-episode run over full-year (8760-step) episodes, seed 0, with the thermal actuators under control.

| Metric                  | Reference controller | STEMS         |
|-------------------------|----------------------|---------------|
| cost                    | 34008.88             | 15219.87      |
| emissions               | 21071.71             | 8688.99       |
| average daily peak      | 36.78                | 16.47         |
| electricity consumption | 142816.61            | 55219.75      |
| ramping rate            | 6.368                | 4.804         |
| discomfort rate         | 0.0001               | 0.0001        |
| safety violation rate   | 0.6465               | <b>0.0000</b> |

Episode reward rose from 205 to 671 between episodes 2 and 15 while the measured violation rate remained at 0.0000 in *every* episode: the safety property held throughout learning, not only at convergence.

**Limitation.** This run predates the hot-water correction described in Section 8, so it is a valid *heat-pump-only* result rather than a joint hot-water and heat-pump one. The planned remedy is Section 9, item 2.

### 6.3 E3 — Hot-water lead time

CityLearn charges the tank by  $e = aC$  but clamps that to the heater’s one-hour output, so the achievable state-of-charge gain in one step is

$$\rho_i^{\text{dhw}} = \min\left(c_i, \frac{P_{\text{nom},i} \eta_i}{C_i}\right), \quad \tau_i = 1/\rho_i^{\text{dhw}} \text{ hours.} \quad (34)$$

Measured across the eight buildings:  $\rho^{\text{dhw}} \in [0.52, 0.85]$  and  $\tau \in [1.17, 1.92]$  h, mean 1.60 h. Equation 34 was validated against the simulator rather than assumed: charging an empty tank at maximum action for one step moves the state of charge to within 15% of  $\rho_i^{\text{dhw}}$  for every building, the residual being tank efficiency and standby loss.

Since  $\tau_i > 1$  h everywhere, a controller that responds to demand *after* observing it is structurally too late. This is the measurement that motivates a pre-heat horizon  $L \geq 2$ .

### 6.4 E4 — Anticipatory pre-heating

1500 steps  $\times$  2 seeds over the heating season. The nominal controller is *price-aware but reactive*: it refills the tank only once demand has appeared and declines to heat above its running-mean price. Measuring against a controller that is already trying to be economical makes the barrier’s contribution interpretable as the value of anticipation specifically. Every configuration is graded by the *same* fixed evaluation barrier, a separate instance from the one being varied.

**Positive result 1: the horizon is set by the physics.** Readiness rises monotonically with  $L$  and the energy deficit falls by 86% between  $L = 1$  and  $L = 2$  (661.7  $\rightarrow$  90.6 kWh), improving only slowly thereafter. This is the discontinuity Eq. 34 predicts: with a mean time-to-heat of 1.60 h a one-hour horizon cannot in general close the gap in time, and a two-hour horizon can. The horizon is determined by measured device physics, not selected by preference – the same lesson as the battery calibration in E1, applied to a different actuator.

Table 3: E4: anticipatory pre-heating. Readiness and coverage higher is better; deficit, cost and \$/kWh lower is better.

| Configuration                                      | Readiness    | Covered      | Deficit [kWh] | Cost   | DHW \$/kWh |
|----------------------------------------------------|--------------|--------------|---------------|--------|------------|
| No pre-heat (reactive)                             | 0.907        | 0.939        | 1026.5        | 3021.6 | 0.2200     |
| + efficiency-aware power guard                     | 0.907        | 0.939        | 1026.5        | 3021.6 | 0.2200     |
| Pre-heat $L = 1$                                   | 0.928        | 0.950        | 661.7         | 3025.9 | 0.2300     |
| Pre-heat $L = 2$                                   | 0.961        | 0.958        | 90.6          | 3033.1 | 0.2411     |
| Pre-heat $L = 3$                                   | 0.989        | 0.968        | 33.8          | 3039.0 | 0.2936     |
| + cold-front anticipation ( $L = 2$ )              | 0.982        | 0.964        | 67.3          | 3035.6 | 0.2517     |
| <b>Full (<math>L = 2</math>, weather + margin)</b> | <b>0.991</b> | <b>0.970</b> | <b>47.4</b>   | 3038.5 | 0.2571     |

**Positive result 2: targeted anticipation is cheaper than uniform conservatism.** Comparing  $L = 3$  against the full configuration ( $L = 2$  plus the cold-front term and a wider margin): the full configuration reaches higher readiness (0.991 vs 0.989) and better coverage while paying 12% less per kWh of hot-water heating (0.2571 vs 0.2936). Extending the horizon raises the requirement at *every* hour; conditioning on the forecast raises it only when the weather signal justifies it, since Eq. 28 is identically zero under steady weather.

**Negative result 1: readiness is not free.** Every configuration that improves readiness also spends more: hot-water electricity rises from 196.2 to 238.4 kWh and the mean price paid per kWh from 0.2200 to 0.2571, lifting total cost by 0.6%. The mechanism differs from E1, where the barrier only ever *restricted* an action the policy already wanted: here the barrier *commands* heating at hours the policy would have skipped, and because the requirement is driven by demand rather than price those hours are on average more expensive than the ones a price-following controller selects. **Planned work:** the barrier currently heats as late as the constraint allows. Selecting *when within the horizon* to pre-heat – the earliest inexpensive hour rather than the last feasible one – should recover most of the premium and turn the barrier into a load-shifting mechanism (Section 9, item 3).

**Negative result 2: the efficiency-aware power guard is inactive.** The first two rows are identical to every reported digit. This is expected rather than faulty: per-building loads are 10–40 kW against an 80 kW cap, so  $h_2$  and  $h_3$  never bind, and a guard that reserves headroom inside a cap that is never approached cannot alter any action. The row is reported rather than omitted so that an inactive mechanism is visible as inactive. **Planned work:** exercised under a tightened cap in E5, and to be evaluated on a heating-dominated climate where the efficiency term is larger (Section 9, item 4).

**Note on the seed spread.** The per-seed standard deviation of readiness is exactly 0.000 throughout. That reflects determinism, not stability: the reactive nominal controller is a deterministic function of the observation and the scenario seed is fixed in the schema, so both seeds trace the same trajectory. These seeds establish reproducibility and say nothing about robustness. **Planned work:** Section 9, item 5.

## 6.5 E5 — Cold-snap stress

Two corrections to the experimental design were required before the scenario tested what it was intended to test, and both are recorded because each is easy to get wrong.

1. **A constant temperature offset cannot exercise a cold-front term.** Equation 28 keys on the *difference* between the current and forecast efficiency. Shifting the current reading and all of its forecasts by the same amount changes the temperature level but leaves that difference essentially untouched. The correction is a forecast-only ramp,  $\hat{T}_{t+k} += \gamma k$  with  $\gamma < 0$ .
2. **The cap must actually bind.** With the battery frozen, mean per-building draw is 0.88 kW; a 12 kW cap is no more binding than the nominal one. The cap had to be reduced to 2 kW before  $h_2$  engaged at all.

Both perturbations act on the observation stream rather than on the simulator’s internal physics, so this scenario probes whether the anticipation logic responds correctly to a weather signal; it is not a statement about true energy use in a cold snap.

Table 4: E5: cold-snap stress ( $-18^\circ\text{C}$  observed,  $-4^\circ\text{C}/\text{h}$  forecast front, 2 kW per-building cap; 1500 steps, seed 0).

| Configuration                                      | Readiness    | Deficit [kWh] | Cost          | DHW \$/kWh | Violation |
|----------------------------------------------------|--------------|---------------|---------------|------------|-----------|
| No pre-heat (reactive)                             | 0.895        | 1163.5        | 3021.6        | 0.2200     | 0.2402    |
| + efficiency-aware guard                           | 0.895        | 1163.5        | <b>3008.2</b> | 0.2200     | 0.2388    |
| Pre-heat $L = 1$                                   | 0.919        | 734.7         | 3013.1        | 0.2307     | 0.2390    |
| Pre-heat $L = 2$                                   | 0.961        | 103.2         | 3021.4        | 0.2459     | 0.2390    |
| Pre-heat $L = 3$                                   | 0.991        | 33.2          | 3028.9        | 0.3127     | 0.2402    |
| + cold-front anticipation ( $L = 2$ )              | 0.986        | 60.2          | 3024.9        | 0.2550     | 0.2390    |
| <b>Full (<math>L = 2</math>, weather + margin)</b> | <b>0.991</b> | <b>43.5</b>   | 3028.3        | 0.2591     | 0.2390    |

**Positive result: anticipation is worth more when there is weather to anticipate.** Adding the cold-front term at  $L = 2$  raises readiness from 0.961 to 0.986 and cuts the deficit by 42% ( $103.2 \rightarrow 60.2$  kWh), a larger gain than the same term produced under steady weather. The full configuration matches the readiness of  $L = 3$  while paying 17% less per kWh (0.2591 vs 0.3127), against 12% under nominal conditions.

**Positive result: the efficiency-aware guard becomes active.** With the cap binding, the second row separates from the first: cost falls 0.44% ( $3021.6 \rightarrow 3008.2$ ), making it the least expensive configuration in the table, because the guard trims conditioning draw hardest exactly when efficiency is lowest.

**Negative result: the guard cannot repair the constraint.** Its effect on the violation rate is negligible ( $0.2402 \rightarrow 0.2388$ ), and the reason quantifies cleanly. With the battery frozen, the shield commands only the hot-water heater and the heat pump, together 366.5 of 10597.3 kWh – **3.5% of total consumption**. The remaining 96.5% is non-shiftable load, and no thermal action can bring a building under a 2 kW cap that its baseline load already exceeds; roughly 24% of (timestep, building) pairs violate  $h_2$  in every configuration because those violations are unavoidable in the sense of Section 5. A shield can only be as effective as the fraction of load it actually commands. **Planned work:** re-enable the battery so the controlled fraction is large enough for the guard to act on, and evaluate the same scenario with vehicles present, where the controllable fraction is far larger (Section 9, item 1).

## 6.6 E6 — Vehicle barrier on real charger data

The vehicle barrier was validated against the shipped CityLearn EV dataset before being used, in the same way the tank calibration was checked against the real tank.

Measurements: a charger rated 11 kW at  $\eta = 0.95$  serving a 60 kWh battery gives  $\rho = 0.174 \text{ h}^{-1}$ , i.e. approximately 5.7 h to fill – roughly three and a half times the hot-water lead time of E3. The test suite asserts that every required observation is present; that  $\rho$  matches  $P_{\text{ch}}\eta/C$  read from the live charger object; that a 72-step rollout spanning genuine arrivals *and* empty bays leaves the barrier finite, in-bounds and inert on empty bays; and that an unreachable departure requirement is reported rather than absorbed.

**Consequence for the horizon rule.** For hot water the horizon is a hyper-parameter validated against a measured time constant. For a vehicle the horizon is supplied by the data itself, so Eq. 22 is exercised with nothing to tune – and the longer lead time makes it correspondingly more consequential.

## 6.7 E7 — The merged building-and-vehicle environment

The merged schema of Section 2.3 was validated end to end. Result: 8 buildings, 6 with chargers, observation dimension 35, action dimension 4. All control modes resolve, and a single rollout driving the hot-water heater and the charger simultaneously moves both stores, confirming that the thermal and vehicle sides are live in the same environment. Charging lead times range from 5.7 to 7.4 h.

The generator asserts that no thermal observation was lost in the merge and that the building data files are referenced, not copied.

## 6.8 E8 — Regression verification

Because the hot-water and heat-pump results are used as a standalone baseline elsewhere, every structural change to the code was verified not to move them. Two levels of check are applied:

- **Operation level.** The pre-refactor hot-water projection is retained as a frozen reference function and differenced against the unified barrier over 200 randomised states  $\times$  3 horizons  $\times$  2 weather settings.
- **End to end.** The complete E4 study is re-run and differenced against the stored results after each structural change.

Across three successive refactors – unifying the barrier, widening the environment for heterogeneous buildings, and adding the merged schema – the maximum relative difference is  $7.359 \times 10^{-9}$ , on the average-daily-peak metric, which is the most summation-order-sensitive quantity reported. That residual is a one-unit-in-the-last-place `float32` effect arising from the order of summation, and is an order of magnitude below the tolerance already present in the readiness comparison. The corresponding test asserts agreement to `float32` precision, not exact equality.

## 6.9 E9 — Coupling sweep with a deadline-locked vehicle fleet

Sections 4.6 argued that per-device feasibility does not survive a shared connection. This experiment measures that directly on the merged environment, and measures what a coordination rule is worth as a function of how tightly the shared constraint binds.

**Setting.** Six vehicles, 55.2 kW of aggregate charger capacity, on the eight-building merged schema. Measured beforehand: the non-vehicle building load alone peaks at 39.6 kW; with every vehicle charging on demand the site peaks at 71.5 kW, and a mean of 4.2 vehicles are connected simultaneously. The shared import cap is swept from 80 kW (slack) to 15 kW (severely binding).

**Arms.** The nominal controller is deliberately uncoordinated: each household charges whatever is plugged in, which is what a distribution transformer sees when a street electrifies. Three shield configurations are compared.

- **independent** – each deadline barrier projects on its own. Correct per device, collectively blind to the connection.
- **proportional** – the shared cap is enforced; every request is scaled by the same factor.
- **edf** – the shared cap is enforced; available power is allocated earliest-deadline-first.

The last two enforce an *identical* total, so any difference between them is attributable to the priority rule alone. All three necessarily coincide wherever the cap is slack, which is what makes a sweep the correct instrument rather than a single operating point.

**Service metric.** A *missed departure* is a vehicle that disconnects with a state of charge below what it required. It is the honest service metric here because, unlike a state-of-charge excursion, it cannot be repaired by any later action.

Table 5: E9: grid-cap sweep, 1500 steps, 298 departures. `miss` is missed departures; `viol` is the fraction of steps exceeding the cap.

| Cap [kW] | independent |       | proportional |       | edf        |       | EDF gain<br>vs. proportional |
|----------|-------------|-------|--------------|-------|------------|-------|------------------------------|
|          | miss        | viol  | miss         | viol  | miss       | viol  |                              |
| 80       | 3           | 0.000 | 3            | 0.000 | 4          | 0.000 | −33%                         |
| 60       | 3           | 0.013 | 1            | 0.000 | 1          | 0.000 | 0%                           |
| 45       | 3           | 0.065 | 6            | 0.001 | 5          | 0.000 | +17%                         |
| 35       | 3           | 0.113 | 20           | 0.009 | <b>11</b>  | 0.011 | + <b>45%</b>                 |
| 30       | 3           | 0.142 | 58           | 0.018 | <b>34</b>  | 0.023 | + <b>41%</b>                 |
| 25       | 3           | 0.179 | 115          | 0.045 | <b>66</b>  | 0.047 | + <b>43%</b>                 |
| 20       | 3           | 0.219 | 170          | 0.099 | <b>115</b> | 0.095 | + <b>32%</b>                 |
| 15       | 3           | 0.273 | 214          | 0.201 | 179        | 0.177 | +16%                         |

**Positive result 1: enforcement converts a breach into a scheduling problem.** The **independent** arm never limits anything: it misses only 3 of 298 departures at every cap, but exceeds the cap in up to 27% of steps and peaks at 73.4 kW regardless of what the cap is set to. Enforcing the shared cap removes almost all of that breach – at 35 kW the violation rate falls from 0.113 to 0.009 – and converts it into a scheduling problem, at the cost of missed departures. This trade is the substance of the coupled-feasibility condition of Eq. 29: when the requirement set cannot be met, something must give, and the only choice available is *what*.

**Positive result 2: the value of the priority rule is non-monotone in coupling tightness.** This is the main finding. Comparing the two arms that enforce the same total:

- At 80 and 60 kW the cap is slack and the rule is worth nothing.
- Between 35 and 20 kW it avoids 32–45% of the missed departures that proportional sharing incurs, peaking at +45% (11 misses against 20) at 35 kW.

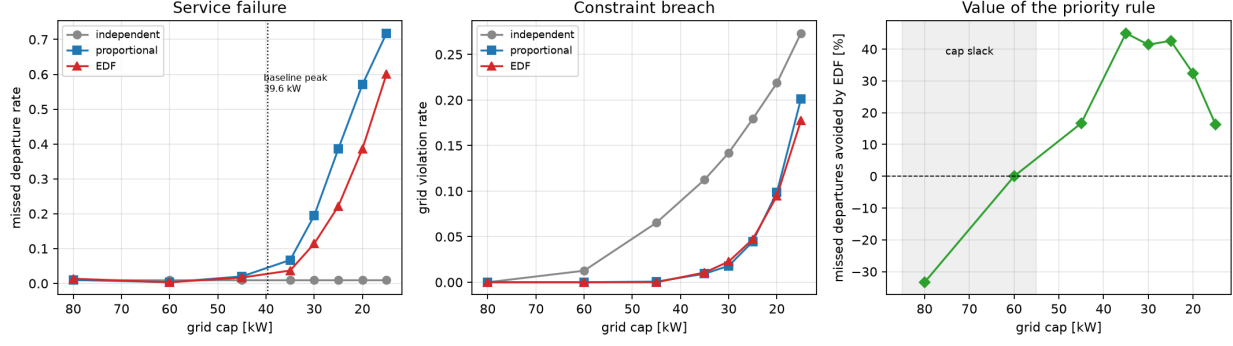


Figure 1: E9. Left: service failure. Centre: constraint breach. Right: the missed departures avoided by earliest-deadline-first relative to proportional sharing, at an identical enforced total. The dotted line marks the 39.6 kW non-vehicle baseline peak, below which no vehicle action can bring the site under the cap.

- At 15 kW the advantage falls back to +16%.

The decline at the tight end is as informative as the peak. Once the cap is far below what the fleet requires, most deadlines are unreachable under any allocation, and a priority rule can only reorder failures rather than prevent them. Coordination is therefore most valuable in an intermediate regime: it is worthless where the constraint does not bind, and its leverage shrinks again where the problem is infeasible for everyone. Reporting a single operating point would have supported almost any conclusion one wished to draw.

**Negative result: a floor set by the uncontrollable load.** Every enforcing arm peaks at exactly 39.6 kW for all caps at or below 30 kW, and the residual violation rate rises regardless of the allocation rule. That figure is the non-vehicle building load, which the shield does not command. Below it, no vehicle action can bring the site under the cap, and the violations are unavoidable in the sense of Section 5. This is the same bound identified in E5 from a different direction, and it quantifies what any charging coordination scheme can and cannot deliver. **Planned work:** repeat with the battery and thermal actuators also under control, which raises the controllable fraction substantially and should lower that floor.

**Negative result: the headroom estimate is conservative.** At 80 kW the three arms should be identical, because the realised peak of 73.4 kW never reaches the cap; instead **edf** records 4 missed departures against 3. The cause is that available headroom is computed from the previously observed net import, which already contains that step’s charging, so the allocator occasionally engages when the cap is in fact slack. The effect is small and conservative in the safe direction, but it is a modelling artifact rather than a property of the system. **Planned work:** estimate the next-step non-controllable load separately from the controllable draw, instead of using the previous total as a proxy.

**Scope.** This experiment characterises the coupling itself and compares *allocation rules inside the safety layer*. It is not a comparison of learned coordination architectures: no policy is trained here, and the nominal controller is fixed by construction so that the shield’s contribution can be isolated. Establishing where the shared constraint binds, and by how much, is a prerequisite for

that comparison – a learned coordinator evaluated at 80kW would have nothing to coordinate about – but it is not a substitute for it. The learned comparison is Section 9, item 1.

## 7 Standalone Heat-Pump Operation and Platform Integration

The hot-water and heat-pump controller is designed to run without the rest of the system. This section documents that mode and sets out what integrating it with a commercial heat-pump platform would require.

### 7.1 Running standalone

Control is restricted by a named registry mapping a mode to a set of actuators, resolved by device name rather than by position:

$$\text{ACTION\_GROUPS} \supseteq \left\{ \text{heatpump}:[\text{heat pump}], \text{dhw}:[\text{DHW}], \text{thermal}:[\text{DHW}, \text{heat pump}] \right\}.$$

Selecting a mode has two effects:

1. **Actuator masking.** Devices outside the selected set are held at their neutral value. For a storage device an action of zero is exactly "neither charge nor discharge" (Eq. 2 with  $e = 0$  leaves the state governed only by standby loss), so freezing them is physically meaningful rather than an arbitrary placeholder.
2. **Barrier adaptation.** The battery state-of-charge barrier is constructed only when the battery is actually controllable, so it is skipped rather than constraining an actuator the controller cannot move. The power barriers remain active in every mode, since hot-water and conditioning draw still affect the building's import.

Requesting a device the installation does not have raises an error rather than silently controlling nothing.

In standalone thermal mode the controller therefore consists of: the observation encoder, the per-building policy, the hot-water readiness barrier  $h_4$ , the efficiency-aware power guard, and the comfort term with its dual-setpoint deadband. No battery, vehicle or neighbourhood coupling is required for it to run.

### 7.2 What the controller needs from a heat-pump installation

The control output is a scalar modulation command per actuator in  $[-1, 1]$ , which maps naturally onto an inverter-driven compressor and onto a hot-water charge request. On-off equipment would require a duty-cycle or hysteresis layer between the barrier output and the device, which is not currently implemented.

### 7.3 Integration path for a commercial platform

Taking a Thermia installation as the concrete case – ground- or air-source units with an inverter-modulating compressor, simultaneous space heating and hot-water production, and a remote monitoring service – integration would proceed in four steps.

**Step 1: commissioning-time calibration.** The single most consequential step, and the lesson of both E1 and E3. The controller must read the installed unit's rated compressor power, tank capacity, heater rating and efficiency, and compute  $\rho^{\text{dhw}}$  and the CoP curve from them. Assuming

Table 6: Signals required for standalone thermal control.

| Quantity                                                            | Cadence                | Use                                                              |
|---------------------------------------------------------------------|------------------------|------------------------------------------------------------------|
| Outdoor temperature, plus a 1–3 h forecast                          | hourly                 | CoP model (Eq. 3); cold-front term (Eq. 28)                      |
| Indoor temperature and set-points (heating and cooling)             | hourly                 | comfort deadband (Eq. 32)                                        |
| Hot-water tank temperature or state of charge                       | hourly                 | barrier state in Eq. 26                                          |
| Hot-water draw (measured or inferred)                               | hourly                 | requirement estimator (Eq. 24); only past values are used        |
| Electrical import                                                   | hourly                 | power barriers $h_2, h_3$                                        |
| Electricity price, ideally day-ahead                                | hourly                 | economic reward term                                             |
| Compressor rated power, tank capacity, heater rating and efficiency | once, at commissioning | calibration of $\rho^{\text{dhw}}$ (Eq. 34) and of the CoP curve |

nominal values across an installed base is precisely the error that produced the highest violation rates in E1. This is a one-time read per installation.

**Step 2: the barrier runs at the edge.** The hot-water barrier (Eq. 26) and the power guard (Eq. 27) are analytic and per-building; they involve no optimisation solver and no neighbourhood state. They can run on the unit controller or a local gateway, at hourly cadence, with negligible computation. This matters for deployment: the safety layer does not depend on connectivity, and it never returns an undefined action – the projection is feasible by construction, and where a requirement is genuinely unreachable it is reported rather than silently approximated.

**Step 3: the demand estimator learns per installation.** The requirement estimator (Eq. 24) is an online hour-of-day climatology that needs no prior training and no data leaving the property; it converges within a few days of observed operation, and during warm-up it falls back to a conservative estimate that never under-reports current demand.

**Step 4: the policy is optional.** The barrier improves hot-water readiness on top of *any* nominal controller – E4 measured its effect on a conventional price-aware reactive controller, not on a learned policy. An installation can therefore adopt the safety and anticipation layer alone, with the learned policy added later. This staging substantially reduces deployment risk.

## 7.4 Recalibration required for a different market

Three items must be revisited before any claim is made about a heating-dominated installed base.

1. **Climate.** All measurements in this report come from a cooling-dominated Texas dataset. The efficiency mechanisms of Section 4.5 were inactive at nominal caps precisely because the climate rarely stresses them (E4, negative result 2). Heating-dominated data is required, and is the planned next step (Section 9, item 4).
2. **Ground-source thermal behaviour.** The CoP model of Eq. 3 is parameterised by a supply and a source temperature. For a ground-source unit the source is the brine circuit, which is far more stable hour-to-hour than outdoor air but drifts seasonally, and over long horizons is

a shared resource between neighbouring installations. Representing that would be a change of source-temperature model, not of the barrier.

3. **Periodic disinfection.** A thermal disinfection cycle is a hard, periodic, regulated requirement that conflicts directly with efficiency, since the elevated tank temperature it needs is expensive to reach. It maps onto the deadline barrier of Section 4.3 without modification: a required state that must be met within a bounded window, at a finite heating rate. This is a natural extension of the existing mechanism and is listed in Section 9, item 7.

## 8 Defects Found and Corrected

Four defects were found during this work. All four are now covered by regression tests. They are recorded here because each one silently affected measurements until it was found.

**D1 — The hot-water actuator was inert.** CityLearn’s storage update for domestic hot water scales the requested energy by the *heating* storage tank’s capacity rather than the hot-water tank’s. The building schema used here has no heating storage tank, so that capacity is zero and every hot-water action resolved to zero energy. Verified directly: commanding a charge action of 0.8 for eight consecutive steps left the tank state of charge at exactly 0.0. The correction is applied per building, only where the defect is present, announced by a banner and recorded in the run metadata, with an explicit opt-out. All device-level hot-water readings taken before this correction are invalid; the affected runs are identified in Section 6.2.

**D2 — Vehicle departure time misread as a clock hour.** The charger schedule field is documented as a countdown in timesteps and the shipped data files decrement it towards zero. An initial implementation of the vehicle barrier treated it as an absolute hour and performed clock arithmetic on it, which would have produced incorrect deadlines at every bay. The field is now used directly. The associated required-state field is stored as a percentage and normalised on load.

**D3 — Departure countdown truncated one step early.** The reference data keeps a vehicle parked while the countdown reaches zero, so the departure hour itself is a final charging opportunity. The schedule generator initially stopped one step earlier, silently removing it.

**D4 — State-of-charge metric scored on a frozen battery.** Under thermal-only control the reported state-of-charge violation rate was pinned at 1.0, because the metric scored the frozen, uncontrolled battery against its band. The metric is now conditioned on whether the battery is actually controllable; the affected mode moved from 1.0 to 0.0.

## 9 Limitations and Planned Work

Each item states an observed limitation and the specific work intended to address it. The ordering reflects how much each would strengthen the conclusions.

1. **Learned coordination has not yet been compared.** E9 measured the shared constraint and compared allocation rules *inside* the safety layer, establishing that the useful operating regime is roughly 20–45 kW on this fleet. It did not train anything. *Planned:* with that regime now known, compare centralised-critic and fully decentralised training across the same sweep,

which is the comparison E9 was built to make possible. Evaluating such a comparison at a slack cap would measure nothing, which is why the sweep came first.

2. **The controllable fraction of load bounds every result.** E5 measured the shield commanding 3.5% of consumption under thermal-only control; E9 found a hard floor at the 39.6 kW non-vehicle baseline. *Planned:* run both with the battery and the thermal actuators under control simultaneously, which is the only way to raise that fraction.
3. **Thermal results predate the hot-water correction.** The converged run of E2 and the earlier control-mode validation drove an inert hot-water actuator (defect D1), so they stand as heat-pump-only results. *Planned:* re-run both to convergence on the corrected environment to obtain a genuine joint hot-water and heat-pump result.
4. **Readiness is bought at a cost premium.** E4 measured a 0.6% increase in total cost and a rise from 0.2200 to 0.2571 in the price paid per kWh of hot-water heating, because the barrier heats as late as the constraint allows and is driven by demand rather than price. *Planned:* choose the heating instant *within* the available horizon rather than at its end, selecting the least expensive feasible hour. Vehicles are the natural first application, because their deadlines leave positive slack whereas hot water does not.
5. **Only one climate has been evaluated.** Every measurement here is from a cooling-dominated dataset, which is why the efficiency-aware mechanisms were inactive at nominal caps. *Planned:* repeat E4 and E5 on the heating-dominated datasets available in the simulator, where the coefficient of performance varies over a much wider range and the cold-front term should carry more information.
6. **Reported spreads are not uncertainty estimates.** The per-seed standard deviation is exactly zero because the nominal controller is deterministic and the scenario seed is fixed. *Planned:* vary the stochastic policy seed, the weather year and the building subset, and report confidence intervals and paired comparisons instead of raw spreads.
7. **Comparative evaluation is incomplete.** The evaluation path currently scores against a single rule-based reference, while several other controllers are implemented but unused. *Planned:* calibrate every one of them from the live device parameters before use – one of them presently carries the same uniform rate assumption identified in E1 – and report the full metric table. Calibrating the comparators is a precondition, not an optimisation.
8. **Periodic disinfection is not yet modelled.** A regulated thermal disinfection cycle is a deadline constraint of exactly the form already implemented. *Planned:* add it as a further instance of the deadline barrier, with the requirement window as the deadline and the elevated tank temperature as the required state.
9. **Vehicle-to-grid capability is unused.** The merged schema is already discharge-capable at 7.2 kW. *Planned:* enable bidirectional operation, which converts a vehicle from a purely constraining load into a flexibility source and changes the structure of the coupled feasibility condition in Eq. 29.
10. **Engineering hygiene.** The project lacks packaging, continuous integration and a container definition, and the requirement estimator’s per-step loop dominates study runtime. *Planned:* package the project, add automated test execution, and vectorise the estimator.

## 10 Summary

This report records a reimplementation of the STEMS architecture on real building data and the extensions built on it. The recurring technical theme is that *actuator dynamics must be measured from the plant rather than assumed*, and that anticipatory safety horizons must be derived from those measurements: a uniform state-of-charge rate left a 9.45% violation rate with the shield active (E1), while per-building calibration and the associated margins reduced the measured rate to zero at no cost penalty; the same principle applied to hot water produced an 86% reduction in energy deficit at exactly the horizon the measured 1.60 h time-to-heat predicts (E4).

The safety layer was then generalised: the hot-water tank, the vehicle battery and the electrical battery are one object – a store that must reach a required state by a deadline at a finite rate – and are implemented once (Section 4.3). That generalisation also exposed the limit of the guarantee: individual barriers are feasible by construction, but coupled deadlines under a shared cap can render the safe set empty (Eq. 29), which the implementation detects and reports rather than absorbing.

The thermal controller runs standalone and is deployable independently of the rest of the system: its barrier is analytic, per-building, solver-free and requires no neighbourhood state, and its demand estimator learns on-site from observed operation (Section 7).

Where mechanisms were inactive or improvements came at a cost, the measurements are reported as observed and paired with the specific work planned to address them (Section 9). Four defects found during the work are documented in Section 8; all are covered by regression tests, and the results they invalidated are identified in place.

## References

- [1] Zhang, X., Wu, J., Zinflou, A., & Boulet, B. (2025). *STEMS: Spatial-Temporal Enhanced Multi-Agent Safe Building Energy Management System*. IEEE Internet of Things Journal. arXiv:2510.14112v2.